

# AMAZON SALES ANALYSIS

## USING POWER BI

Professional Business Intelligence Project Report

|                                                             |                                                          |
|-------------------------------------------------------------|----------------------------------------------------------|
| <b>Domain</b><br>E-Commerce / Amazon Product Analytics      | <b>Technology</b><br>Microsoft Power BI                  |
| <b>Analysis Focus</b><br>Price, Discount, Ratings & Reviews | <b>Visualization</b><br>Interactive Analytical Dashboard |

### Executive Summary

This report presents a Power BI analysis of an Amazon sales dataset. The dashboard focuses on product-level pricing and discount information together with customer rating and review-related metrics. Interactive visuals make it easier to compare products, categories, ratings, and review activity and provide a foundation for data-driven e-commerce decisions.

Prepared as an academic/business intelligence project

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# 1. PROJECT OVERVIEW

## 1.1 Introduction

Amazon is a large-scale e-commerce environment where product information, pricing, discounts, ratings, and reviews generate valuable analytical data. Power BI can transform this information into dashboards that help users understand product performance and customer engagement.

## 1.2 Problem Statement

A large product dataset can be difficult to interpret through raw tables. Business users need visual summaries to compare discount levels, prices, rating counts, ratings, and product categories. The objective of this project is to organize these measures into an understandable Power BI report.

## 1.3 Objectives

- Analyze discount percentage in relation to actual product price.
- Compare product-level pricing and discount patterns.
- Analyze rating counts across product categories.
- Study rating and review-related measures.
- Create interactive Power BI visualizations for business analysis.
- Generate insights that can support product and customer-engagement decisions.

## 1.4 Dataset Fields Represented in the Dashboard

| Field / Measure     | Purpose in Analysis                                               |
|---------------------|-------------------------------------------------------------------|
| product_id          | Identifies individual products.                                   |
| actual_price        | Represents the actual/listed product price used in the dashboard. |
| discount_percentage | Shows the discount percentage associated with products.           |
| category            | Groups products for category-level comparison.                    |
| about_product       | Product-description field used in the rating-count visual.        |
| rating              | Represents product rating information.                            |
| rating_count        | Represents the number/count of ratings.                           |
| review_id           | Identifier used for review-level aggregation in the dashboard.    |

## 2. POWER BI DASHBOARD

### 2.1 Dashboard Snapshot

The following screenshot is the Power BI dashboard supplied for this report. It contains four major analytical visuals covering discount percentage and actual price, rating and rating count, category-level rating counts, and rating by review identifier.

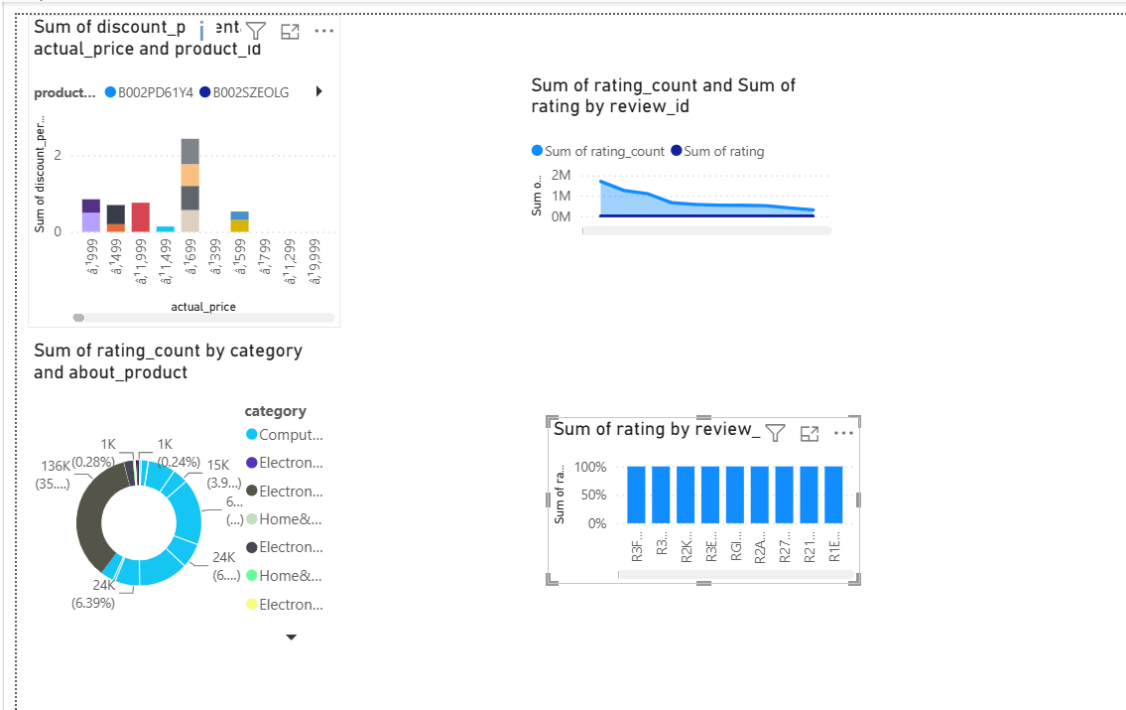


Figure 1. Amazon Sales and Product Analytics Dashboard in Power BI

### 2.2 Dashboard Components

- **Discount Percentage vs. Actual Price:** compares discount percentage across actual-price ranges and selected product identifiers.
- **Rating Count and Rating by Review ID:** displays rating-count and rating measures across review identifiers.
- **Rating Count by Category and Product Information:** summarizes rating counts across product categories and associated product information.
- **Rating by Review ID:** compares the aggregated rating measure across review identifiers.

### 3. ANALYSIS & BUSINESS INSIGHTS

#### 3.1 Price and Discount Analysis

The upper-left visual compares the sum of discount percentage with actual-price ranges and product identifiers. The chart shows that discount activity is distributed across multiple price ranges, with some ranges contributing more strongly than others. This type of analysis can help identify where promotional pricing is concentrated.

#### 3.2 Rating Count by Category

The donut chart provides a category-level view of rating counts. The visible dashboard indicates that rating activity is not evenly distributed across all categories. Some category segments make a larger contribution to the total rating count, indicating stronger customer interaction in those segments.

#### 3.3 Rating Count and Rating by Review ID

The line/area visual compares rating count and rating measures by review identifier. The rating-count series shows a downward pattern across the displayed review identifiers, while the rating measure remains comparatively close to the lower portion of the scale. This demonstrates the difference between review volume and the rating value itself.

#### 3.4 Rating by Review ID

The lower-right chart presents rating values across selected review identifiers. The bars appear uniform on the displayed percentage scale, indicating that the visual is showing an aggregated or normalized measure rather than directly comparing raw rating counts.

#### 3.5 Key Business Insights

| Area                 | Business Interpretation                                                            |
|----------------------|------------------------------------------------------------------------------------|
| Discounts            | Discount activity can be compared across product-price ranges.                     |
| Customer Engagement  | Rating count provides an indicator of customer interaction with products.          |
| Category Performance | Category-level rating counts help identify segments with stronger review activity. |
| Reviews              | Review-level analysis can be used to monitor rating and engagement patterns.       |
| Decision Support     | Interactive Power BI visuals make product and customer metrics easier to compare.  |

*Note: The observations above are based on the supplied Power BI screenshot. Exact numerical conclusions should be taken directly from the underlying Power BI dataset/model.*

## 4. CONCLUSION & FUTURE SCOPE

### 4.1 Conclusion

The Amazon Sales Analysis project demonstrates how Power BI can be used to transform e-commerce data into an interactive business intelligence report. The dashboard combines product pricing, discount percentage, ratings, rating counts, categories, and review-related measures in a single analytical view.

The analysis provides a practical foundation for understanding promotional activity and customer engagement. Instead of relying only on raw records, decision-makers can use the dashboard to compare products and categories and identify patterns that may require further investigation.

### 4.2 Recommendations

- Use product and category filters to investigate high and low rating activity.
- Compare discount percentage with actual price before evaluating promotional effectiveness.
- Monitor rating counts as an indicator of customer engagement.
- Combine ratings with sales quantity and revenue for a stronger product-performance analysis.

### 4.3 Future Scope

- Add total sales, quantity sold, revenue, and profit measures.
- Develop monthly and yearly sales trend analysis.
- Add customer segmentation and category performance dashboards.
- Create Top-N product and best-seller analysis.
- Implement sales and demand forecasting using machine learning.
- Connect Power BI to a live or regularly refreshed Amazon dataset.
- Add AI-based recommendations and automated business insights.

### 4.4 Technologies Used

**Microsoft Power BI** – dashboard and visualization

**Power Query** – data cleaning and transformation

**DAX** – measures and analytical calculations

**Amazon Sales Dataset** – source e-commerce data

**Data Modeling** – organization of analytical fields

### Final Outcome

The completed Power BI report provides a professional analytical framework for Amazon product and customer-engagement data. It can be extended into a comprehensive e-commerce intelligence solution by adding revenue, profitability, forecasting, customer analytics, and AI-driven insights.